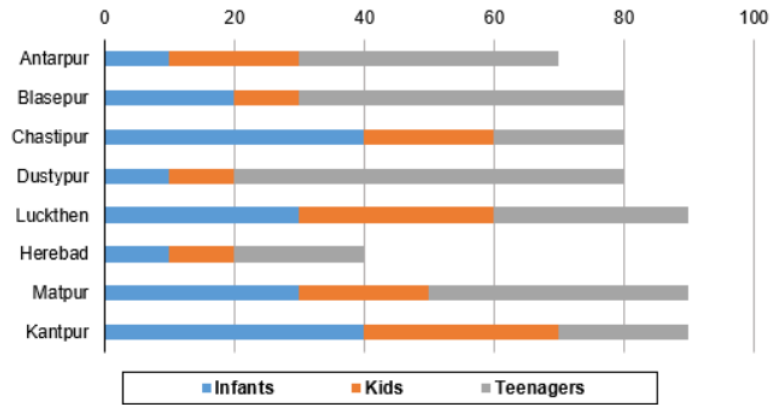


DIRECTIONS for questions 1 to 5: Answer these questions on the basis of the information given below.

The population of each of eight cities – Antarpur, Blasepur, Chastipur, Dustypur, Luckthen, Herebad, Matpur and Kantpur – was divided into exactly five groups – Infants, Kids, Teenagers, Adults and Elderlies. The following graph provides the number (in mn) of Infants, Kids and Teenagers in each of the eight cities:



It is also known that the total number of kids and adults is the same across the eight cities. Further, the total number of teenagers, adults and elderlies is also the same across the eight cities.

Q1. DIRECTIONS for question 1: Select the correct alternative from the given choices.

Which city has the highest population?

- ☐ a) Antarpur
- ☒ b) Kantpur ✓ Your answer is correct
- ☐ c) Matpur
- ☐ d) Either Kantpur or Matpur

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

If the number of adults in Antarpur is two-thirds the number of adults in Dustypur, what is the total number of adults (in mn) in all the eight cities combined?

Q3. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

If the number of adults in Herebad is twice the number of elderlies in Chastipur, which of the following can best be said about the ratio of the population of Matpur to that Kantpur?

- ☐ a) It will range from 0.867 to 1.
- ☐ b) It will range from 0.854 to 1.
- ☐ c) It will range from 0.963 to 1.
- ☒ d) It will range from 0.913 to 1. ✓ **Your answer is correct**

Q4. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

If the number of adults in Herebad is the same as the number of elderlies in Kantpur and the number of adults in Antarpur is twice the number of elderlies in Blasepur, what is the population (in mn) of Luckthen?

- ☐ a) 210
- ☐ b) 270
- ☒ c) 240 ✓ **Your answer is correct**
- ☐ d) 290

Q5. DIRECTIONS *for questions 3 to 5:* Select the correct alternative from the given choices.

If the number of adults is the same as the number of elderlies in at least two cities, which of the following can best be said about the ratio of the populations of Chastipur to Dustypur?

- ☐ a) It will be between 1 and 1.333.
- ☐ b) It will be between 0.967 and 1.333.
- ☐ c) It will be between 0.967 and 1.
- ☐ d) It will be between 1 and 1.4.

DIRECTIONS *for questions 6 to 10: Answer these questions on the basis of the information given below.*

In a college, there were a certain number of students, and each student took one or more among five subjects – OM, OB, ECON, FRA and MAC. However, any student who took MAC also took FRA, while any student who took OM also took OB. It is also known that each student took at least three of the five subjects.

It is also known that

- i. the number of students who took exactly three subjects was the same as the number of students who took both FRA and ECON.
- ii. the ratio of the number of students who took OM and that who took ECON is 9: 10.
- iii. the number of students who took exactly four subjects is 80 and those who took exactly five subjects is 100.
- iv. among the students who took OM, exactly 130 students also took MAC and among the students who took MAC, exactly 125 students took ECON.
- v. not more than 155 students took MAC and, among the students who took FRA, exactly 230 students also took OB.

Q6. DIRECTIONS *for question 6: Type in your answer in the input box provided below the question.*

What is the total number of students in the college?

Q7. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

What BEST can be said about the number of students who took ECON but not OB?

- ☐ a) **Exactly 10**
- ☐ b) **Either 10 or 15**
- ☐ c) **Exactly 20**
- ☐ d) **Exactly 15**

Q8. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

Which of the following provides the comprehensive list of subjects that would have been taken by the maximum number of students?

- ☐ a) **OM, OB**
- ☐ b) **FRA**
- ☐ c) **ECON**
- ☐ d) **OB**

Q9. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.
What BEST can be said about the total number of students who did not take only ECON?

- ☐ a) **Either 30 or 35**
- ☐ b) **Either 25 or 30**
- ☐ c) **Exactly 30**
- ☐ d) **Exactly 25**

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

What is the difference between the number of students who took exactly three subjects and the number of students who took MAC?

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

In the year 2018, three universities, IIS, IIB and IIK, together conducted a Research Aptitude Test (RAT). The following information is known to all the candidates who appeared for the RAT:

- The scores of RAT were valid for a period of five years (from 2018 to 2022) and during these five years, the test would be conducted only once, which was at the beginning of 2018.
- The candidates who appeared for RAT can apply to their preferred university in any year, from 2018 to 2022. However, each candidate can apply only once, irrespective of the university that he applies to.
- Each university has a certain total number of vacant seats to fill over the above-mentioned five-year period (provided in Table 1) from among the candidates who appeared for RAT.
- Among the candidates who apply to any university in any year, the university will select the candidates who achieved the highest marks in RAT and will select the maximum possible number of candidates, subject to the availability of vacant seats in that year.

Exactly fifteen candidates, A through O, appeared for RAT in 2018. Further, when the results were declared, an administrative error led to the scores of all the fifteen candidates (provided in Table 2) being shared with each candidate. Also, each of the fifteen candidates had a first preference and a second preference regarding the university that he wanted to join (also provided in Table 2). The preferences of each candidate were not known to the other candidates.

Every year, each candidate will

- first logically infer as much information as possible, based on the facts available to him.
- apply to any university if and only if that university is either his first or second preference and he is sure that he will be accepted into that university in that year.
- apply to the university which is his second preference, only if he is sure that there are no vacancies in the university which is his first preference.

Each candidate knows that every candidate would follow the above process for applying to any university. Further, at the end of each year, every candidate knew the exact university that each candidate had joined during the year.

Table 1

University	Total Seats Available during the five-year period
IIB	6
IIK	6
IIS	3

Table 2

**Sorted by RAT Score*

Candidate	RAT Score	First Preference	Second Preference
E	99.5	IIB	IIS
F	99.2	IIB	IIK
B	99.1	IIS	IIK
L	98.7	IIS	IIB
O	98.6	IIB	IIK
A	98.4	IIS	IIB
D	98.2	IIK	IIB
I	98.1	IIS	IIB
M	98.0	IIK	IIS
G	97.5	IIS	IIK
J	97.2	IIB	IIS
N	97.0	IIB	IIS
K	96.8	IIB	IIK
C	96.5	IIB	IIS
H	96.3	IIK	IIB

Q11. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

In which year(s) did the maximum number of students join any university?

- ☐ a) **2018**
- ☐ b) **2019**
- ☐ c) **2020**
- ☐ d) **Both 2018 and 2019**

Q12. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

Who among the following did not join any university in the year 2020?

☐ a) **N**

☐ b) **K**

☐ c) **C**

☐ d) **H**

Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

How many students joined a university that was their second preference?

☐ a) 4

☐ b) 2

☐ c) 5

☐ d) 3

Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

What is the total number of seats that were vacant at the beginning of 2022 across all the universities?

☐ a) 0

☐ b) 1

☐ c) 2

☐ d) 3

Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

In which year did H join any university?

☐ a) 2020

☐ b) 2019

☐ c) 2022

☐ d) 2021

DIRECTIONS for questions 16 to 20: Answer these questions on the basis of the information given below.

Four persons, A through D, own all the shares of five different companies – C1 through C5. The number of shares (in mn) of the five companies are 20, 50, 80, 100 and 120, not necessarily in the same order. In each of the five companies, the percentage shares of the four persons are distinct positive multiples of 5.

It is also known that

- i. the number of shares of C1 that A owns is the same as that of C4 that A owns.
- ii. C owns 25% of the shares of C4, and the number of shares of C4 that C owns is the same as the number of shares of C3 that A owns.
- iii. B owns exactly 5 mn shares in each of C2, C3 and C4; the percentage share of B in C3 is the same as that of D in C4 and the percentage share of B in C4 is the same as that of D in C5.
- iv. the number of shares of C5 that C owns is the same as the number of shares of C1 that B owns; the number of shares of C2 that C owns is the same as the number of shares of C1 that D owns.
- v. B owns a greater number of shares in C1 as compared to C5, while A owns an equal number of shares in C2 and C5.

Q16. DIRECTIONS for questions 16 and 17: Type in your answer in the input box provided below the question.

How many shares (in mn) of C1 does B own?

Q17. DIRECTIONS *for questions 16 and 17:* Type in your answer in the input box provided below the question.

What is the difference (in mn) between the number of shares of C2 that A owns and the number of shares of C1 that C owns?

Q18. DIRECTIONS *for questions 18 and 19:* Select the correct alternative from the given choices.

Who owns the maximum number of shares of C5?

☐ a) **B**

☐ b) **C**

☐ c) **A**

☐ d) **D**

Q19. DIRECTIONS *for questions 18 and 19:* Select the correct alternative from the given choices.

Which of the following is definitely true of the number of shares of C3 that D owns?

- ☐ a) It is less than the number of shares of C5 that A owns.
- ☐ b) It is greater than the number of shares of C5 that D owns.
- ☐ c) It is less than the number of shares of C1 that C owns.
- ☐ d) It is greater than the number of shares of C2 that B owns.

Q20. DIRECTIONS *for question 20:* Type in your answer in the input box provided below the question.

What is the difference (in mn) between the total number of shares (across the five companies) that A owns and the total number of shares (across the five companies) that B owns?